KEVIN SCHWARZWALD

Email: kevin.schwarzwald@columbia.edu • Phone: +1 310 961 6126 • ORCID • GitHub: ks905383 • Website • Scholar

Interdisciplinary climate scientist helping inform climate impacts projections and decision-making under uncertainty

EDUCATION

Columbia University — New York, NY
Doctor of Philosophy, Earth and Environmental Sciences — October 2024
Master of Philosophy, Earth and Environmental Sciences — October 2023
Master of Arts, Earth and Environmental Sciences — May 2021
Research Areas: climate variability, uncertainty in projections of climate and climate impacts, precipitation in East Africa, tropical convection, climate model evaluation, the use of climate projections in policy and adaptation efforts
Advisors: Lisa Goddard (†2022), Richard Seager; *Committee Members*: Mingfang Ting, Kate Marvel, Radley Horton
- PhD thesis: *The Present and Future of the Horn of Africa Rains*. Link.

Peking University, Yenching Academy (北京大学燕京学堂) — Beijing, China
Master of Law in China Studies (concentration in Politics and International Relations) — July 2017
Honors: Awards for Outstanding Academic Achievement and Outstanding Contributions to the Academy
- Master's thesis: *The Relationship between Fiscal Systems and Inefficient Urban Expansion in the PRC: An Empirical Look at Prefecture-Level Urbanization*

The University of Chicago — Chicago, IL
Bachelor of the Arts (A.B.) in Physics; Public Policy (concentration in Economics) — June 2015
Honors: Graduated with General Honors, Dean's List 2011-2013, 2015, Departmental Honors in Public Policy
- Senior Thesis (Public Policy): *"The Wrong Folks Kept Winning": Protectionism and America's Trains*
- Study Abroad: Paris, France

ACADEMIC RESEARCH EXPERIENCE

University of California, Davis, Department of Environmental Science and Policy — September 2026 - present
Postdoctoral Scholar — New York, NY (remotely)
PIs: Fran Moore, Adam Sobel, and Michael Tippett
- Studying the Value of Information of climate projections

Columbia Climate School — September 2024 – August 2026
Climate School Postdoctoral Scientist — New York, NY
Mentors: Radley Horton and Gernot Wagner
- Projects included: understanding and constraining the range of climate uncertainty in projections of climate impacts; the suitability of Global Warming Levels in regional climate impacts studies; improving forecast-based financing methodologies for anticipatory humanitarian actions

International Research Institute for Climate and Society (IRI), Lamont-Doherty Earth Observatory (LDEO) — September 2019 – August 2024
Graduate Research Assistant — New York, NY
- Projects included: the present and future of rainfall dynamics in East Africa, understanding the sources of climate uncertainty in projections of the societal impacts (agriculture, mortality, etc.) of climate change, and building tutorials and python tools to make climate data more accessible to end users

Energy Policy Institute at the University of Chicago (EPIC) — September 2017 – July 2019
Pre-Doctoral Fellow (joint with RDCEP) — Chicago, IL and remotely from New York, NY as GISS Visiting Researcher
Principal Investigator: Amir Jina
- Projects included: adapting latest downscaling and bias-correction methods to impacts studies, understanding how changes in climate variability impacts common economic damage
- Advised on the use of climate model and observational data in economics and policy applications

Center for Robust Decision-Making on Climate and Energy Policy (RDCEP) — April 2015 – July 2019
Pre-Doctoral Fellow (joint with EPIC 2017-2019) — Chicago, IL and remotely from New York, NY, and Beijing, China
Principal Investigator: Elisabeth Moyer
- Projects included: understanding changes in precipitation and temperature variability from synoptic to inter-decadal timescales in global climate models

PUBLICATIONS (upcoming)

Schwarzwald, Kevin, Nathan Lenssen, Tom Bearpark, Alia Bonanno, Radley Horton, and Gernot Wagner. "Observational uncertainty is the primary source of climate uncertainty in impacts projections" *Under review at Science Advances.*

Schwarzwald, Kevin, Nathan Lenssen, Radley Horton, and Gernot Wagner. "A bias-corrected & downscaled massive ensemble to diagnose uncertainty in climate impact projections." *Under revision at Scientific Data.* Link.

Mauerman, Max, **Kevin Schwarzwald**, Daniel Osgood, Nathan Lenssen, and Abdel-Lathif Younous. "Sampling Uncertainty is Critical for Setting Forecast-Based Anticipatory Action Triggers" *Under revision at Bulletin of the American Meteorological Society.*

Lenoble, Colin, Mathias Valla, Jonas Schmitt, Céline Guivarch, Améline Vallet, Pauline Rivoire, **Kevin Schwarzwald**, and Kai Kornhuber. "Global risks of renewable energy droughts under climate change" *Under revision at Nature Communications.*

Schwarzwald, Kevin and Nathan Lenssen. *Solicited textbook chapter, in progress.* "The Sources of Uncertainty in Climate Projections." in *Encyclopedia of Climate System Science*, edited by Sumant Nigam and Isla R. Simpson. Elsevier Major Reference Works. Elsevier.

Kisembe, Jesse, Yixin Wen, Chris Funk, **Kevin Schwarzwald**, Ronald I. Odongo, Weikang Qian, and Samuel Ekwacu. "Asymmetric Indo-Pacific Drivers Produce Later Shorter Seasons with Earlier Longest Dry Spells or Earlier Longer Seasons with Later Longest Dry Spells." *Submitted to Journal of Hydrometeorology.*

PUBLICATIONS (peer-reviewed)

Schwarzwald, Kevin and Kerrie Geil. 2024. "`xagg`: A Python Package to Aggregate Gridded Data onto Polygons." *Journal of Open Source Software* 9 (104): 7239. Link.

Schwarzwald, Kevin and Richard Seager. 2024. "Revisiting the 'East African Paradox': CMIP6 models also struggle to reproduce strong observed MAM drying trends." *Journal of Climate* 37 (24): 6641-58. Link.

Rising, James A., Azhar Hussain, **Kevin Schwarzwald**, and Ana Trisovic. 2024. "A Practical Guide to Climate Econometrics: Navigating Key Decision Points in Weather and Climate Data Analysis." *Journal of Open Source Education* 7 (75): 90 Link.

Schwarzwald, Kevin, Richard Seager, Mingfang Ting, and Alessandra Giannini. 2023. "Large-Scale Stability and the Greater Horn of Africa Long and Short Rains." *Journal of Climate* 36 (20): 7297–7317. Link.

Anderson, Weston, Benjamin I. Cook, Kim Slinski, **Kevin Schwarzwald**, Amy McNally, and Chris Funk. 2023. "Multiyear La Niña Events and Multiseason Drought in the Horn of Africa." *Journal of Hydrometeorology* 24 (1): 119–31. Link.

Schwarzwald, Kevin, and Nathan Lenssen. 2022. "The Importance of Internal Climate Variability in Climate Impact Projections." *Proceedings of the National Academy of Sciences* 119 (42). Link.

Schwarzwald, Kevin, Lisa Goddard, Richard Seager, Mingfang Ting, and Kate Marvel. 2022. "Understanding CMIP6 Biases in the Representation of the Greater Horn of Africa Long and Short Rains." *Climate Dynamics* 61 (3): 1229–55. Link.

Schwarzwald, Kevin, Andrew Poppick, Maria Rugenstein, Jonah Bloch-Johnson, Jiali Wang, David McInerney, and Elisabeth J. Moyer. 2021. "Changes in Future Precipitation Mean and Variability across Scales." *Journal of Climate* 34 (7): 2741–58. Link.

AWARDS, GRANTS, COMPETITIVE PROGRAMS

- Climate School Postdoctoral Program; successful institutional fellowship application for "*The Full Magnitude of Climate Uncertainty in Projections of Climate Impacts: A Case Study in Tropical, Developing Countries*", **$145,430** (salary) — 2024-2026
- CATER School on Transdisciplinary Climate Risk Adaptation and Action (Naivasha, Kenya; fully funded, including travel) — 2023
- ASciNA Young Scientist Award, **$7500** — 2023
- Outstanding Oral Presentation Award, 103rd AMS Annual Meeting — 2023
- Center for Science and Society Seed Grants, **$3975** total (to NENSIC, co-PI) — 2021, 2022, 2023
- ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics (Trieste, Italy; fully funded) — 2022
- Yenching Academy Dean's Research Grant, **¥5000** — 2016

OTHER EXPERIENCE

New Era Network for Societally Integrated Climatology (NENSIC) May 2020 – present
Co-founder, Organizer *Website*: nensic.org

- Launched an interdisciplinary network designed to connect young researchers, students, and professionals working on climate issues in any field, with a focus on improving interdisciplinary communication, building informal connections, peer mentoring, and providing a welcoming, inclusive venue to present work and find collaborators
- Organized and co-organized 60+ events in person and remotely with participants from four continents, with research talks on topics such as international climate negotiations, writing public comments as a scientist, or the coordination of disaster planning in Ecuador, in addition to networking events, tutorials, and interdisciplinary job panels

Eparque Urban Strategies March 2019 – July 2019
Independent Consultant (part-time) New York, NY

Yenching Global Symposium (全球青年中国论坛) October 2015 – April 2016
Associate Director, Logistics Beijing, China

Technology Policy Institute July 2014 – August 2014
Google Policy Fellow Washington, DC

Paris Center for Cosmological Physics January 2014 – March 2014
Lab Intern Paris, France

alpS GmbH August 2013 – September 2013
Management Intern, Consulting Group Innsbruck, Austria

Moser/Jaritz August 2013 – September 2013
Research Intern Gmunden, Austria

INVITED AND CONTRIBUTED PRESENTATIONS

2026: International Institute for Applied Systems Analysis (IIASA), NYU Institute for Policy Integrity, Mailman School of Public Health, EGU General Assembly (**highlighted talk**)
2025: MLISTRAL2 Meeting (**invited keynote**), American Geophysical Union (AGU) Fall Meeting
2024: JPL Science Understanding through Data Science (SUDS) Seminar; European Geosciences Union (EGU)'s General Assembly, Max Planck Institute for Meteorology Hamburg, Oxford School of Geography and the Environment
2023: American Meteorological Association (AMS)'s Annual Meeting; EGU General Assembly; University of California Santa Barbara (UCSB) Climate Hazards Center; ECS Symposium; WCRP Open Science Conference; AGU Fall Meeting (**invited abstract**)
2022: AGU Fall Meeting; Institute for Mathematical and Statistical Innovation (IMSI); NASA Goddard Institute for Space Studies (GISS); ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics; Interdisciplinary PhD Student Workshop in Sustainable Development (IPWSD); Columbia Student Climate Symposium
2021: AGU Fall Meeting; University of Chicago Environmental Data Science Lunch
2019: US CLIVAR Large Ensembles Workshop
2018: AGU Fall Meeting

DATA TOOLS AND TUTORIALS

Weather Panel Tutorial climateestimate.net

- In an inter-disciplinary team, co-developed a tutorial designed to introduce climate economists to working with observational climate data using python (xarray), R, or MATLAB

xagg github.com/ks905383/xagg

- Lead developer and maintainer of **xagg**, a python package for aggregating raster data (such as gridded weather data) onto polygons (such as administrative shapefiles), using area-averaging based on the overlap between pixels, polygons

quantproj github.com/ks905383/quantproj

- Developer of **quantproj**, an R package for projecting weather data by using downscaled fine-scaled distributional changes in Large Ensembles and the delta method; based on the methodology of Haugen et al., 2018

SKILLS

Analysis: advanced proficiency in python (xarray, dask), MATLAB, R, and Excel, basic proficiency in GIS (QGIS) and Stata, experience with climate (NetCDF, zarr, nco, CMIP, etc.) and satellite data analysis (DMSP/OLS, VIIRS, etc.)

High-performance computing: proficiency with work on HPC systems, including working with schedulers, writing optimized and parallelized code, etc., proficiency with NCAR's HPC and file storage system
Modeling: experience in running SPEEDY, an intermediate-complexity global climate model (GCM)
Communication/Design: advanced proficiency in LaTeX, PowerPoint, Keynote, inkscape; basic proficiency in Adobe Illustrator
Language: native in German, fluent in French, basic communication skills in Mandarin Chinese

PROFESSIONAL NETWORKS

- American Geophysical Union (AGU)
- European Geosciences Union (EGU)
- Austrian Scientists & Scholars in North America (ASciNA)